Lecture 16

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Midterm 2

- Midterm 2 will be in-class next Wednesday, November 11
- Midterm 2 covers probability:
  - Chapters 13 - 17 in the text,
  - Lectures 9 - 15 and associated review exercises,
  - Quizzes 5-7, and
  - R material from lectures and labs related to probability and simulation.

R Code on Midterm 2

- Midterm 2 may ask for R code.
  - Calculate binomial or hypergeometric probabilities.
  - Simulate random variables.
  - Estimate probability, expected value, or standard error by simulation.
  - Use normal approximation to estimate probability.

Lecture 16

1. Reading Summary - Chapter 18
2. The Glivenko-Cantelli Theorem
3. The Central Limit Theorem
4. Using the Normal Approximation

Read Chapters 19 and 20 for next time.

Quiz 7 is on Wednesday in lecture.

Reading Summary - Chapter 18

Section 18.1: Introduction

- A little history
- The probability histogram of the binomial distribution closely follows the normal curve.

Section 18.2: Probability Histograms

- A **probability histogram** represents chance by area.
- An **empirical histogram** is a data histogram, where the data comes from a finite number of identically simulated random variables.
- As the number of simulated random variables increases, the empirical histogram converges to the probability histogram.
  - This is called the **Glivenko–Cantelli theorem**, which is not in the book.

Section 18.3: Probability Histograms and the Normal Curve

- Suppose you toss a coin many times and let X be the number of heads.
  - The probability histogram of X approximately follows the normal curve.
  - The approximation gets better and better with more flips.

Section 18.4: The Normal Approximation

- Detailed example of using the normal approximation to compute the probability of getting a certain number of heads
- Continuity correction - applies to integers

Section 18.5: The Scope of the Normal Approximation

- The normal approximation applies to more than coin flipping.
- It is useful for the mean of draws in any box model, as long as the number of draws is large enough.
- This is called the **Central Limit Theorem**.
  - How many draws are needed?
  - No clear answer - it depends on the contents of the box.
  - Lopsided boxes require more draws.
  - The Central Limit Theorem applies to means of draws.
  - Not products, etc.

Section 18.6: Conclusion

- If the probability histogram follows the normal curve, then the expected value and standard error tell you everything you need to know about the distribution.
- You can estimate probabilities in this case by converting to standard units like before.

The Glivenko–Cantelli (GC) Theorem

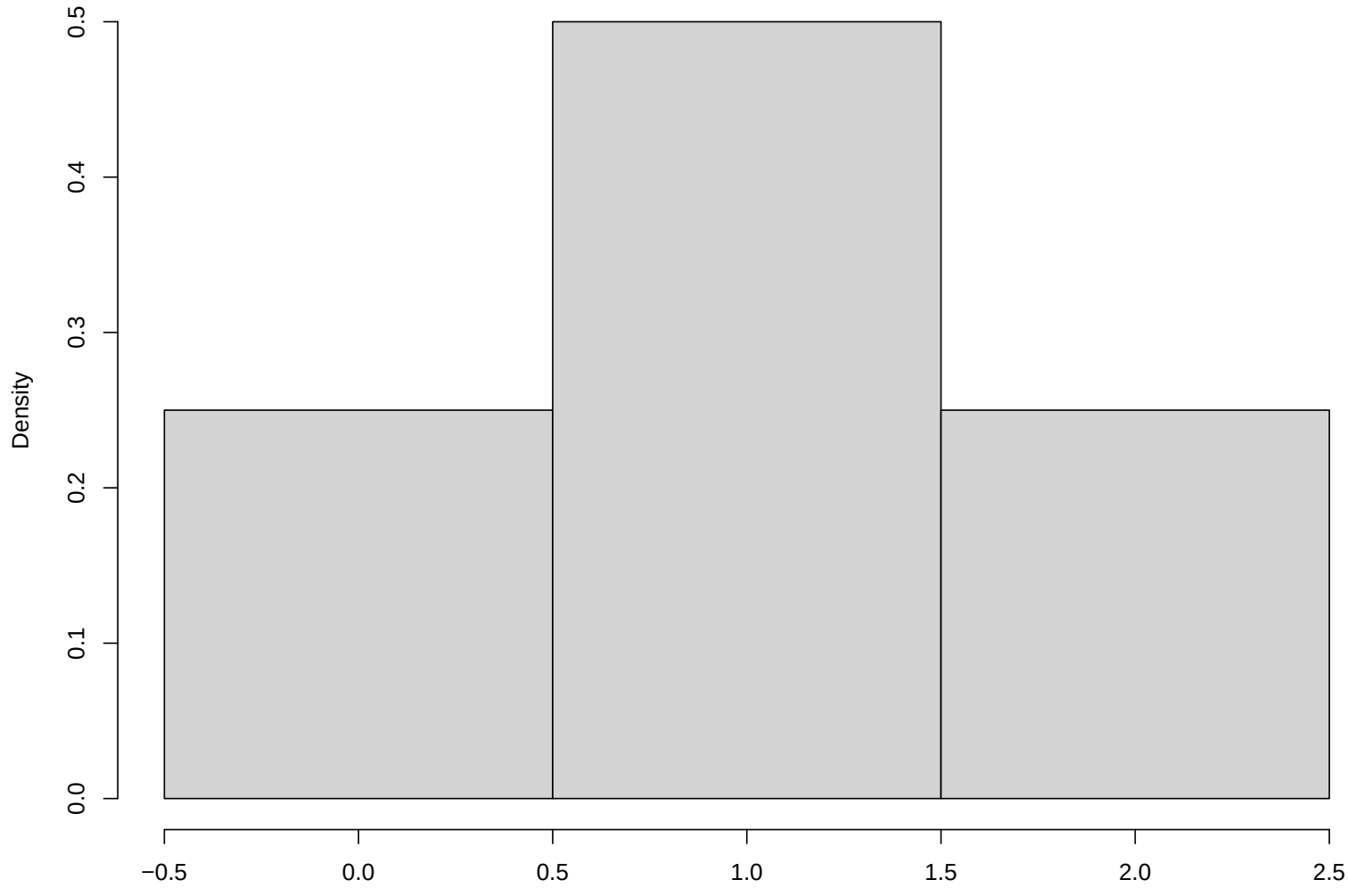
- Suppose we have a sequence of **independent, identically distributed** (IID) random variables X_1, X_2, X_3, ..., X_n.
  - Think draws with replacement from a box.
- The **empirical histogram** is the data histogram of X_1, X_2, X_3, ..., X_n.
- The Glivenko–Cantelli Theorem states that the empirical histogram converges to the probability histogram with more and more draws.
- Nothing really surprising here!

Histograms for Integer Data

```
# This function makes histograms that look nice
# with integer data
int_hist = function(x)
    {
    options(repr.plot.width=5.5, repr.plot.height=5)
    par(mar=c(4,4,1,1))
    hist(x, breaks=seq(min(x)-0.5, max(x)+0.5, by=1),
         freq=FALSE, main="", xlab="")
}
```

Example: Flipping Two Coins

- Suppose you flip two coins and let X be the number of heads.
- Probability Histogram

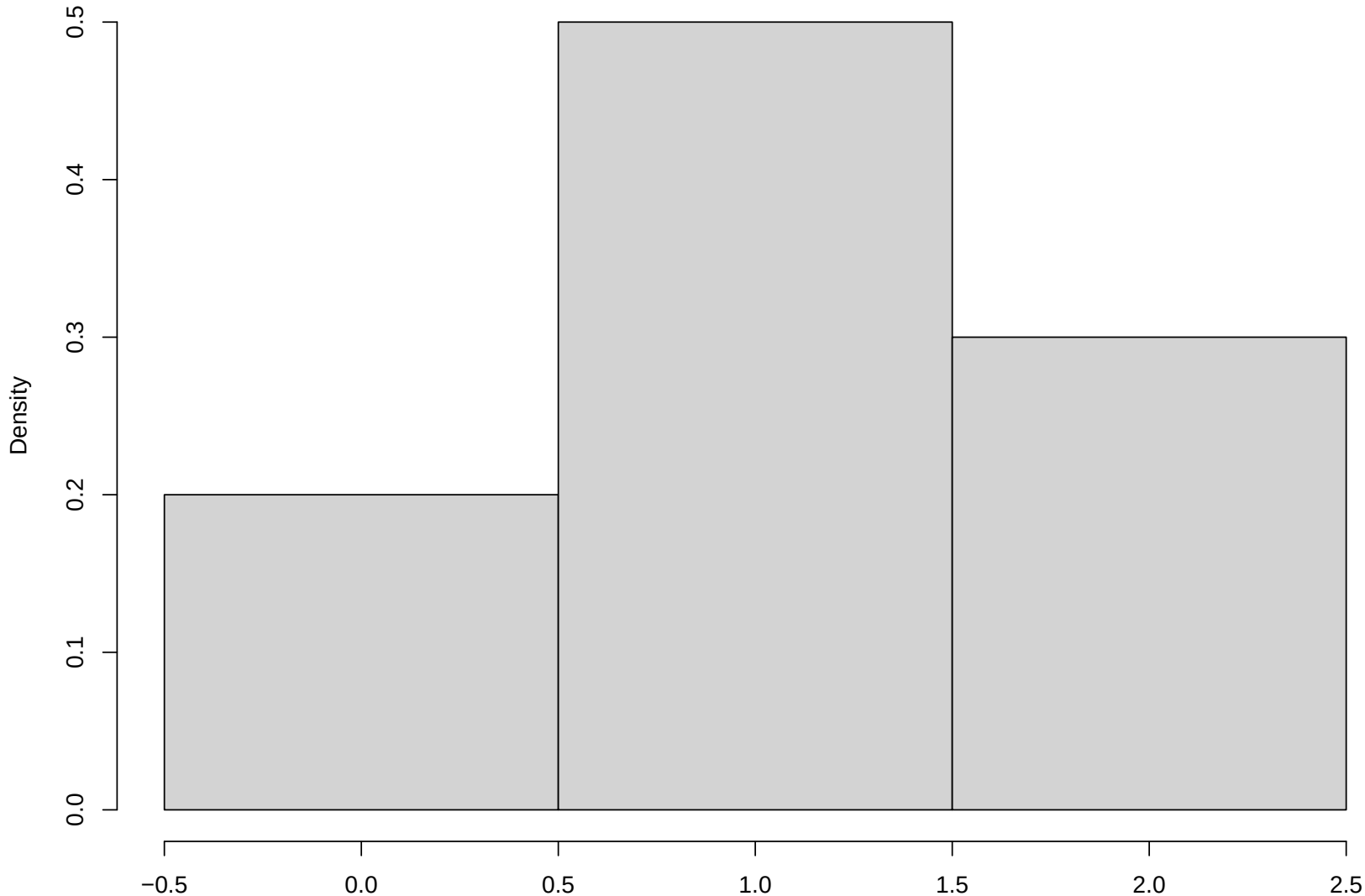


Simulating Two Flips Many Times

```
#Simulates n repetitions of two coin flips
twocoins = function(n) {
  p = runif(n)
  return(0*(p<1/4)+1*(p>=1/4 & p<3/4)+2*(p>=3/4))
}
```

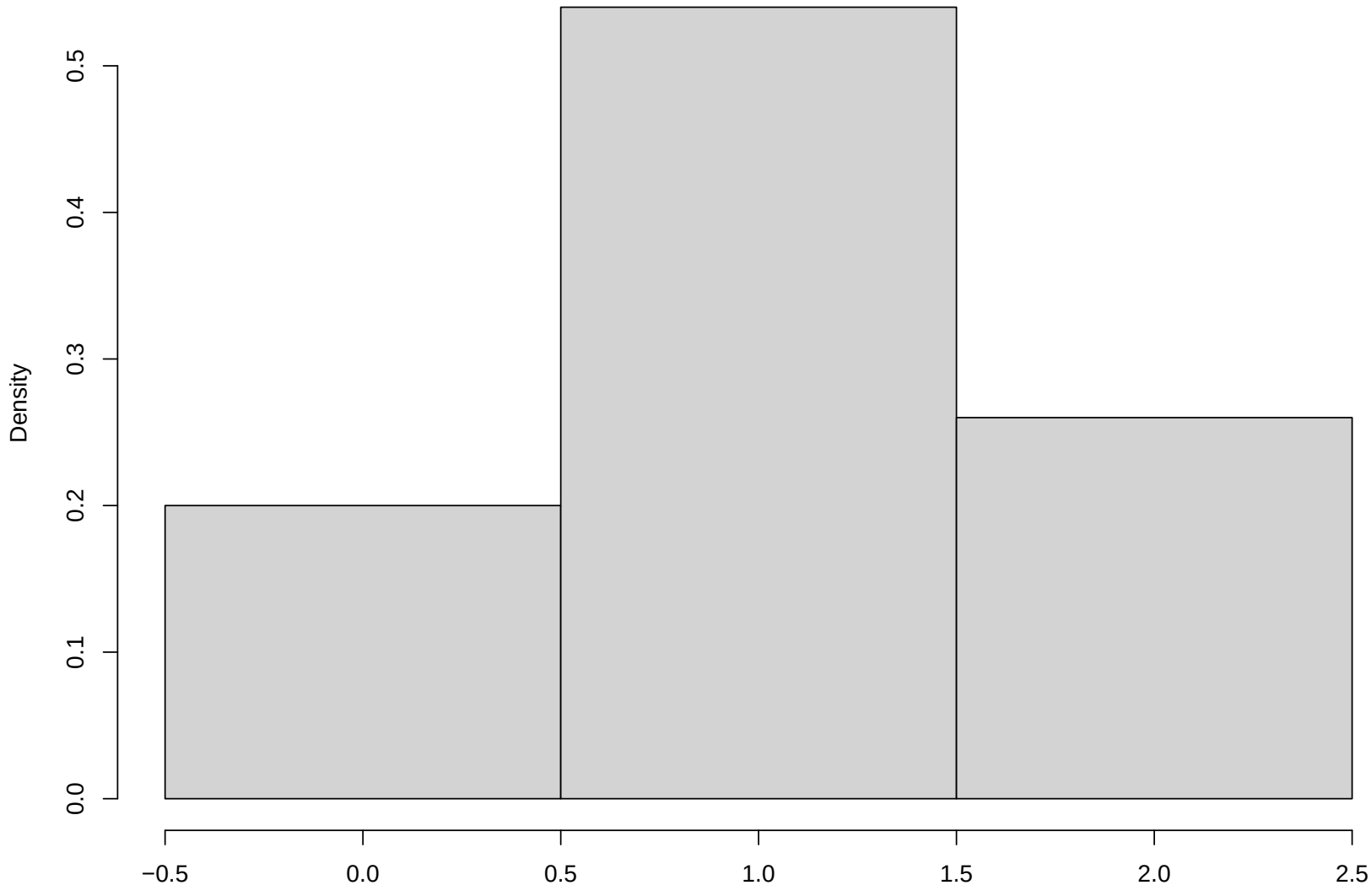
Data Histogram for 10 Repetitions of Two Flips

```
set.seed(1) #Ensures replicability of results
int_hist(twocoins(10))
```



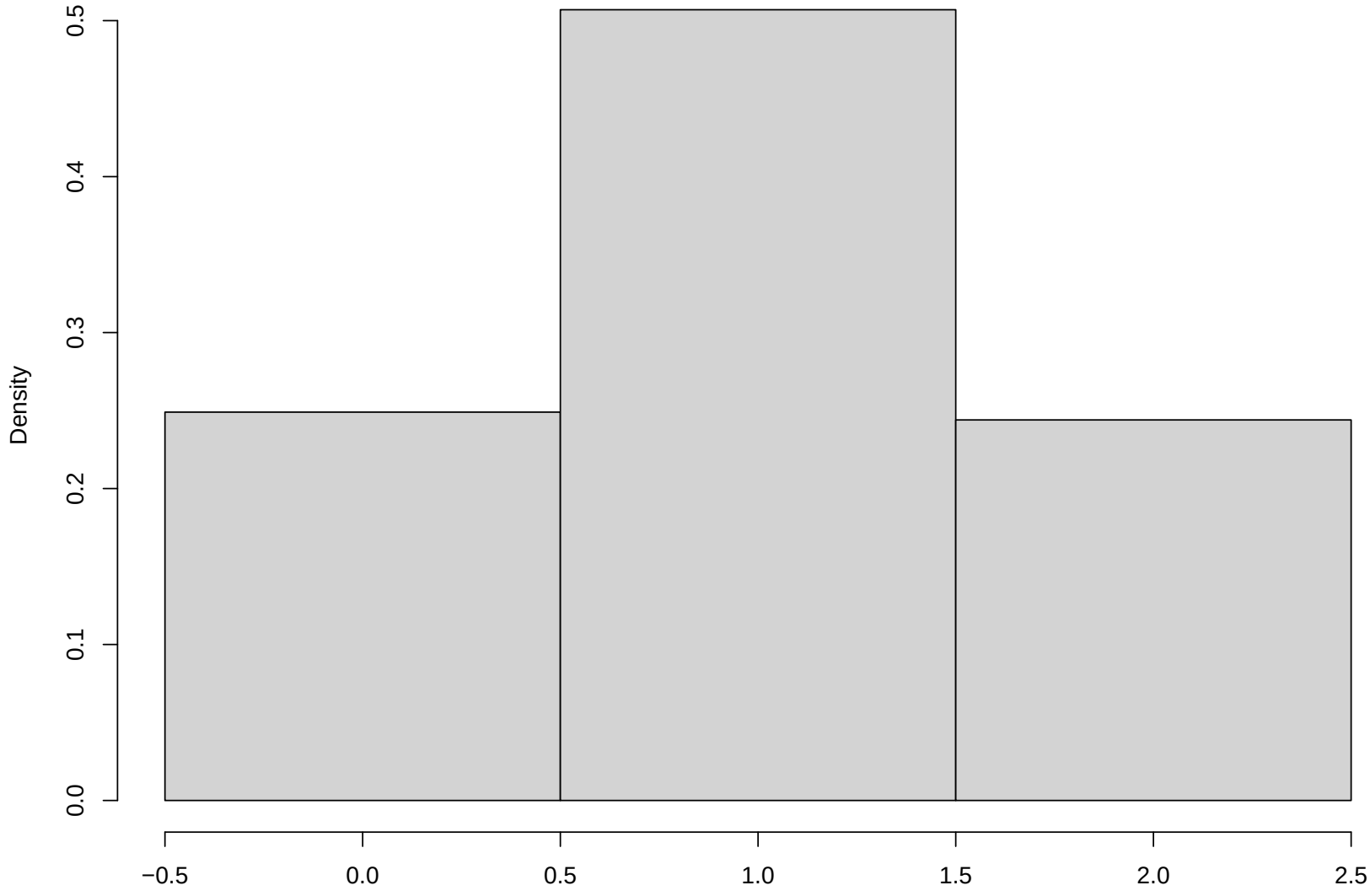
Data Histogram for 100 Repetitions of Two Flips

```
int_hist(twocoins(100))
```



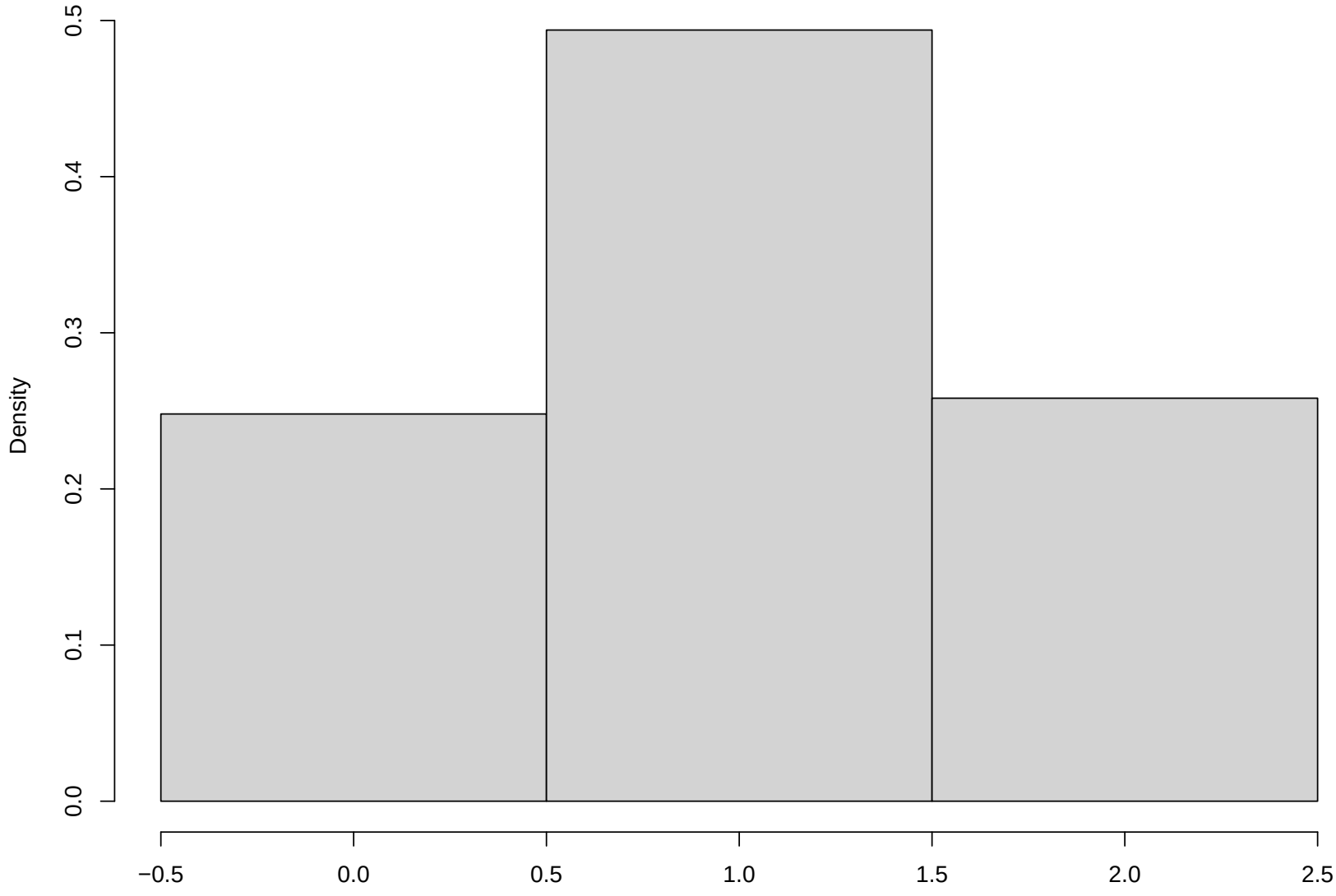
Data Histogram for 1,000 Repetitions of Two Flips

```
int_hist(twocoins(1000))
```



Data Histogram for 10,000 Repetitions of Two Flips

```
int_hist(twocoins(10000))
```

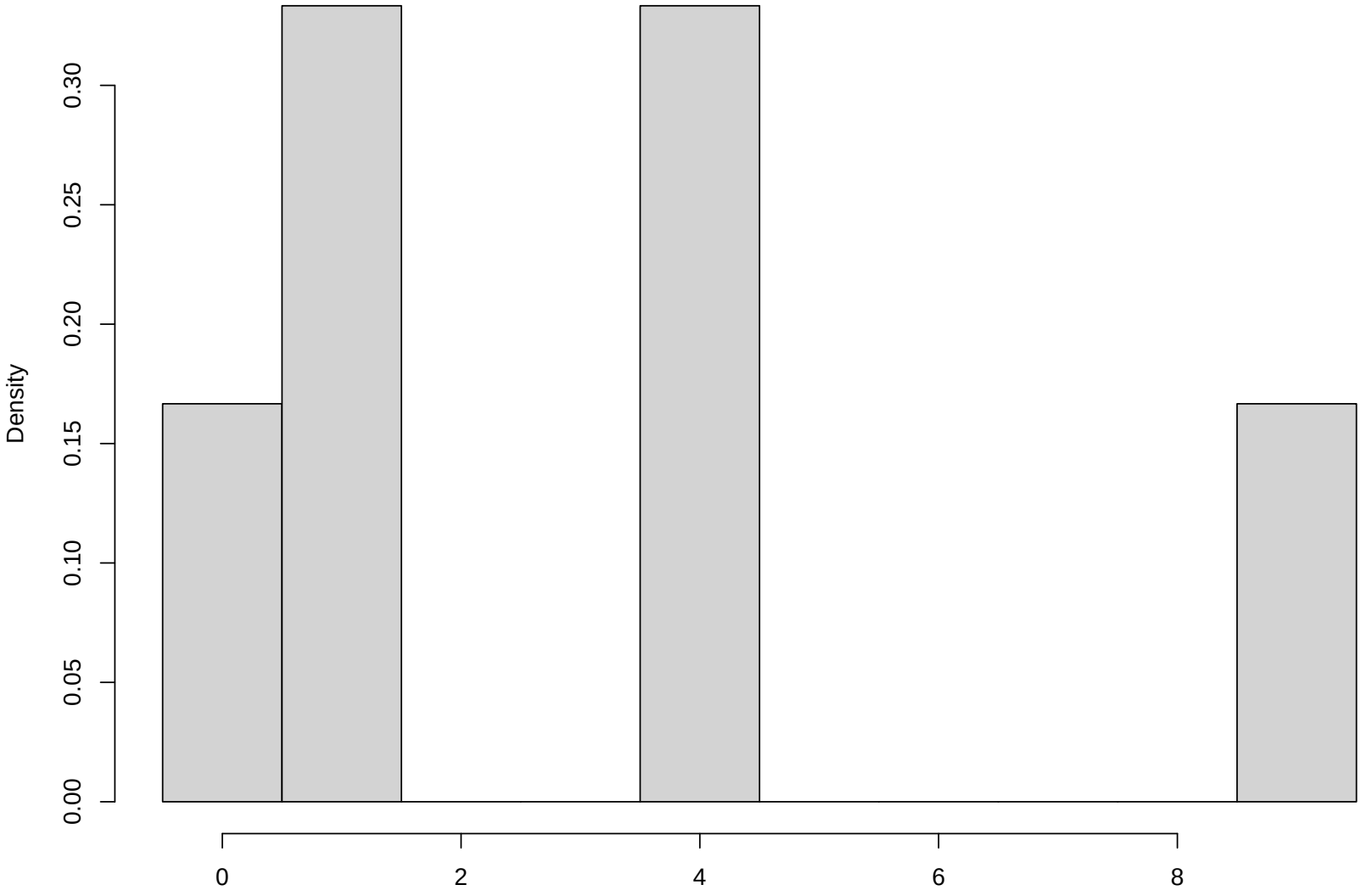


A More Complicated Random Variable

- Let X be the number on a die and $Y = (X - 3)^2$.

x	P(X=x)	Y
1	1/6	4
2	1/6	1
3	1/6	0
4	1/6	1
5	1/6	4
6	1/6	9

Probability Histogram of Y



The sample Function

- The sample function in R allows us to sample multiple times with replacement from the given vector:

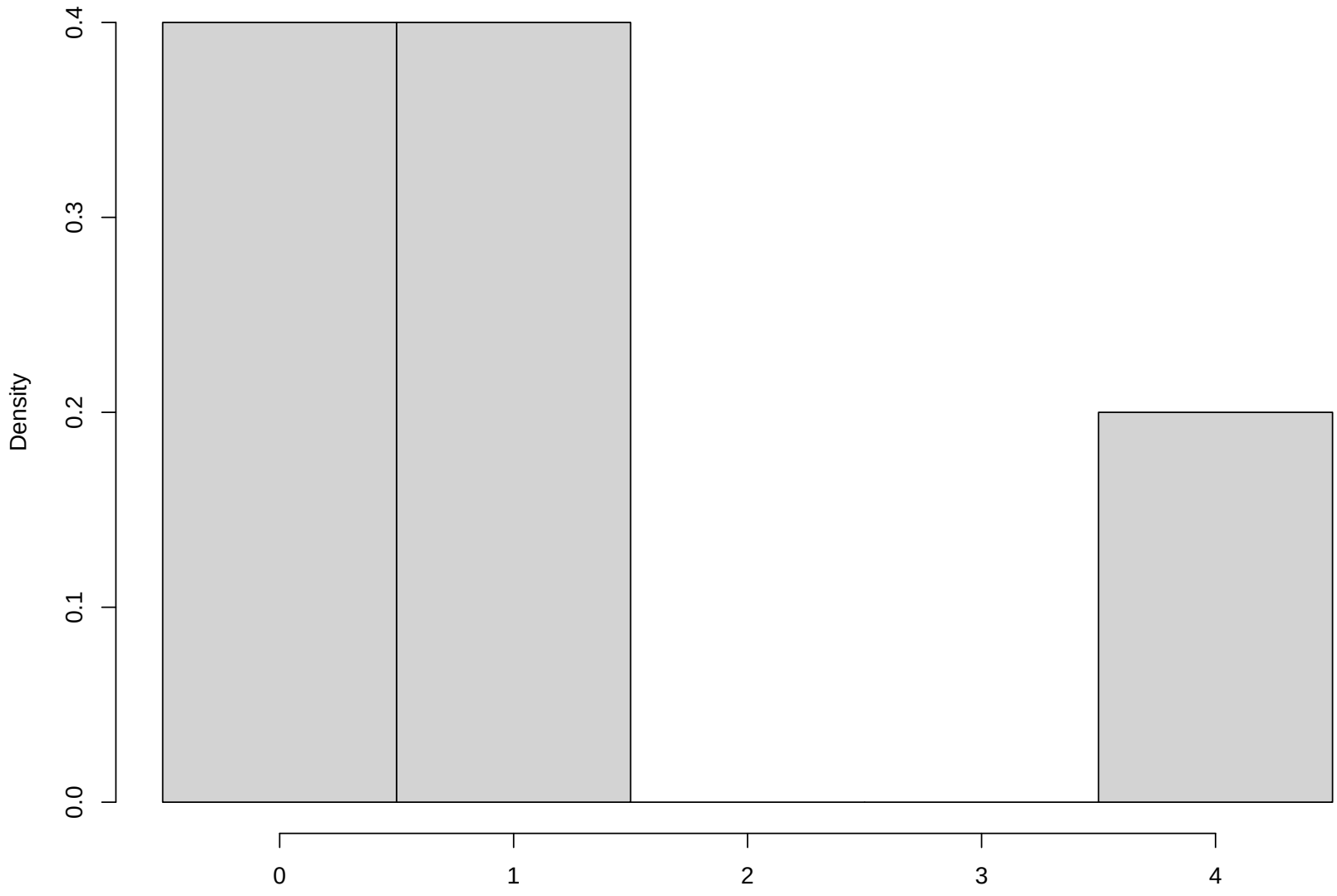
```
#Simulates 1000 die rolls with replacement
X = sample(c(1,2,3,4,5,6),1000,replace=TRUE)
```

Simulating Y

```r
#Simulates n independent values of Y
sim_y=function(n) {
  X = sample(c(1,2,3,4,5,6),n,replace=TRUE)
  return((X-3)^2)
}
```

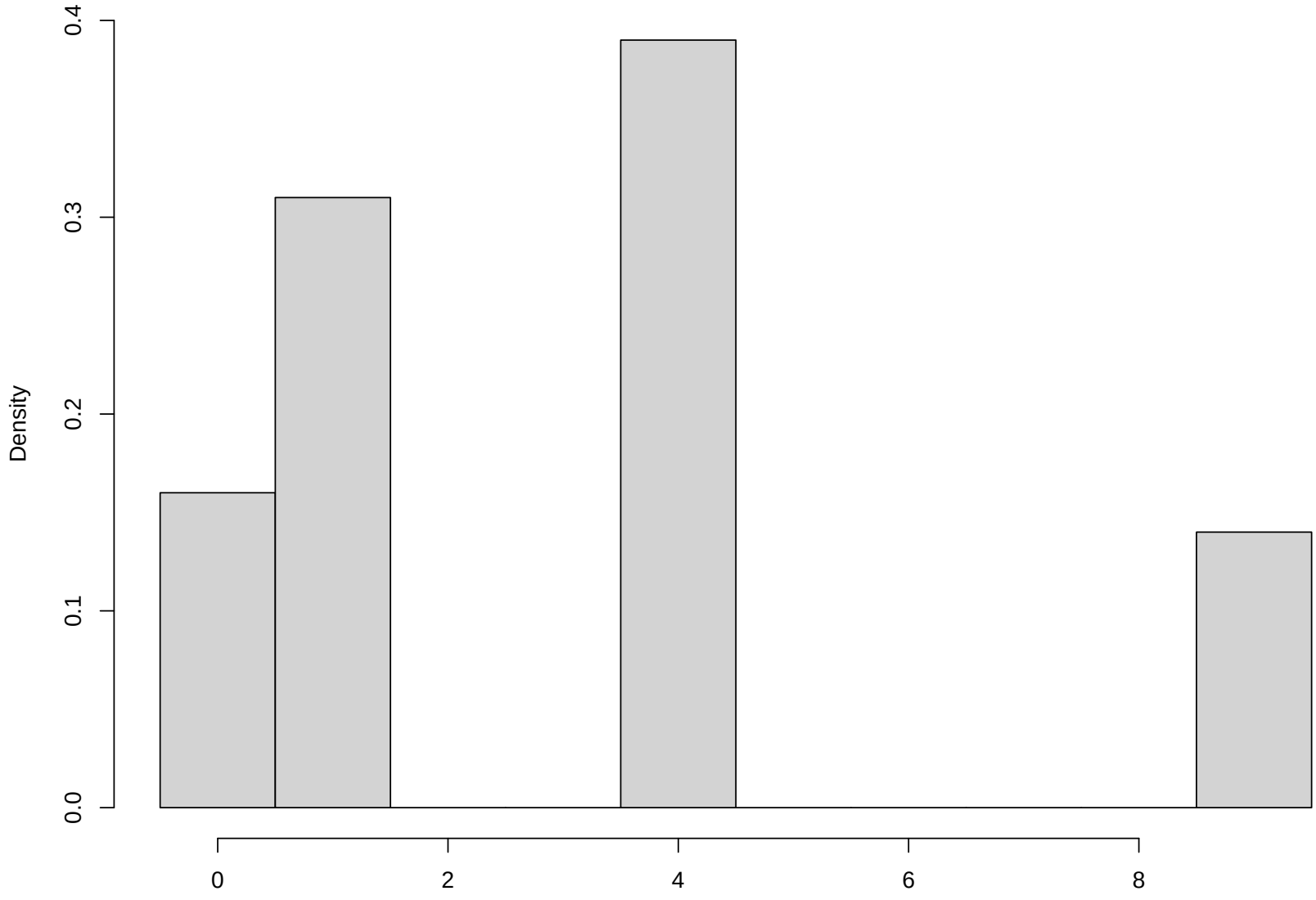
Data Histogram for 10 Draws of Y

```
int_hist(sim_y(10))
```



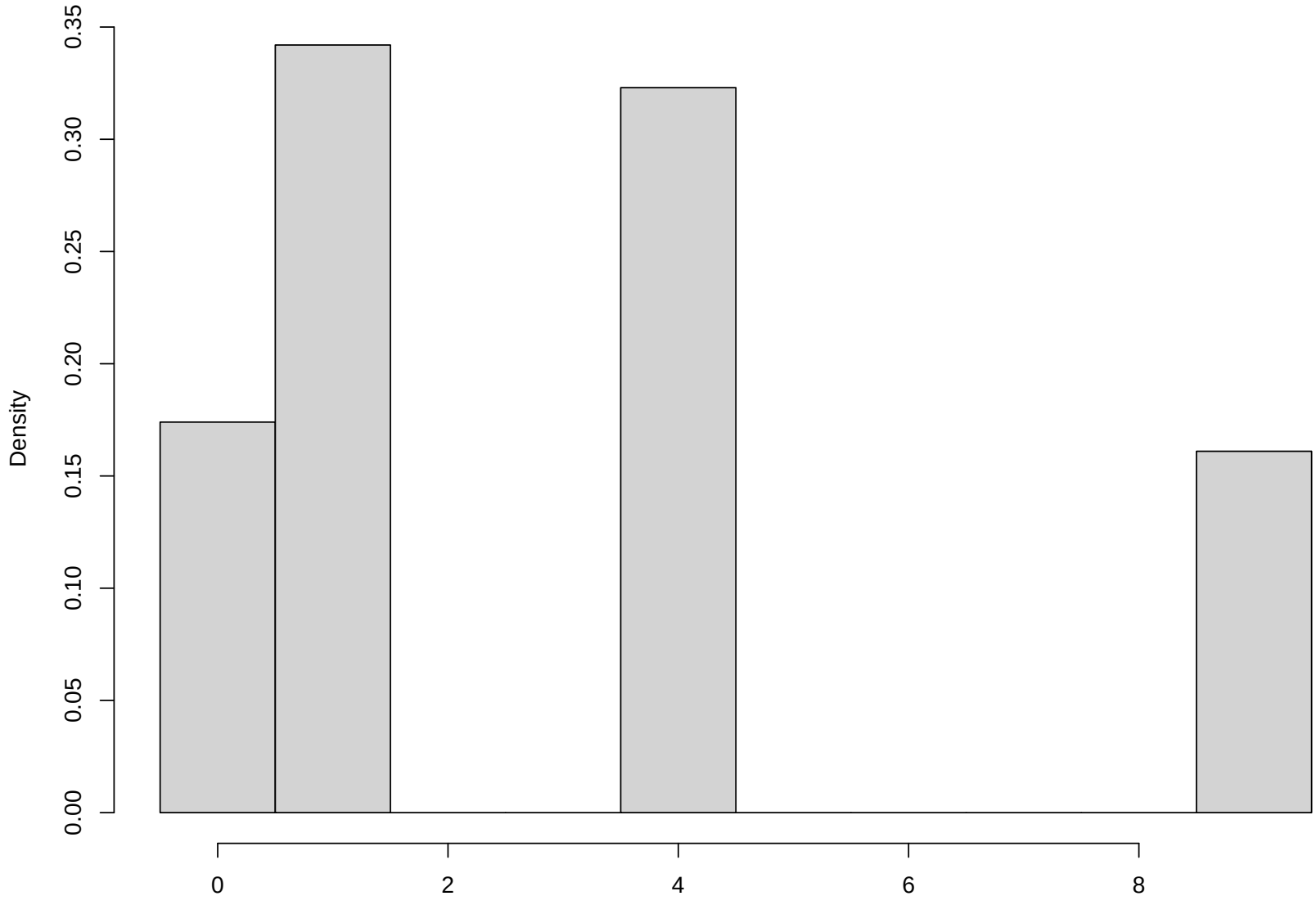
Data Histogram for 100 Draws of Y

```
int_hist(sim_y(100))
```



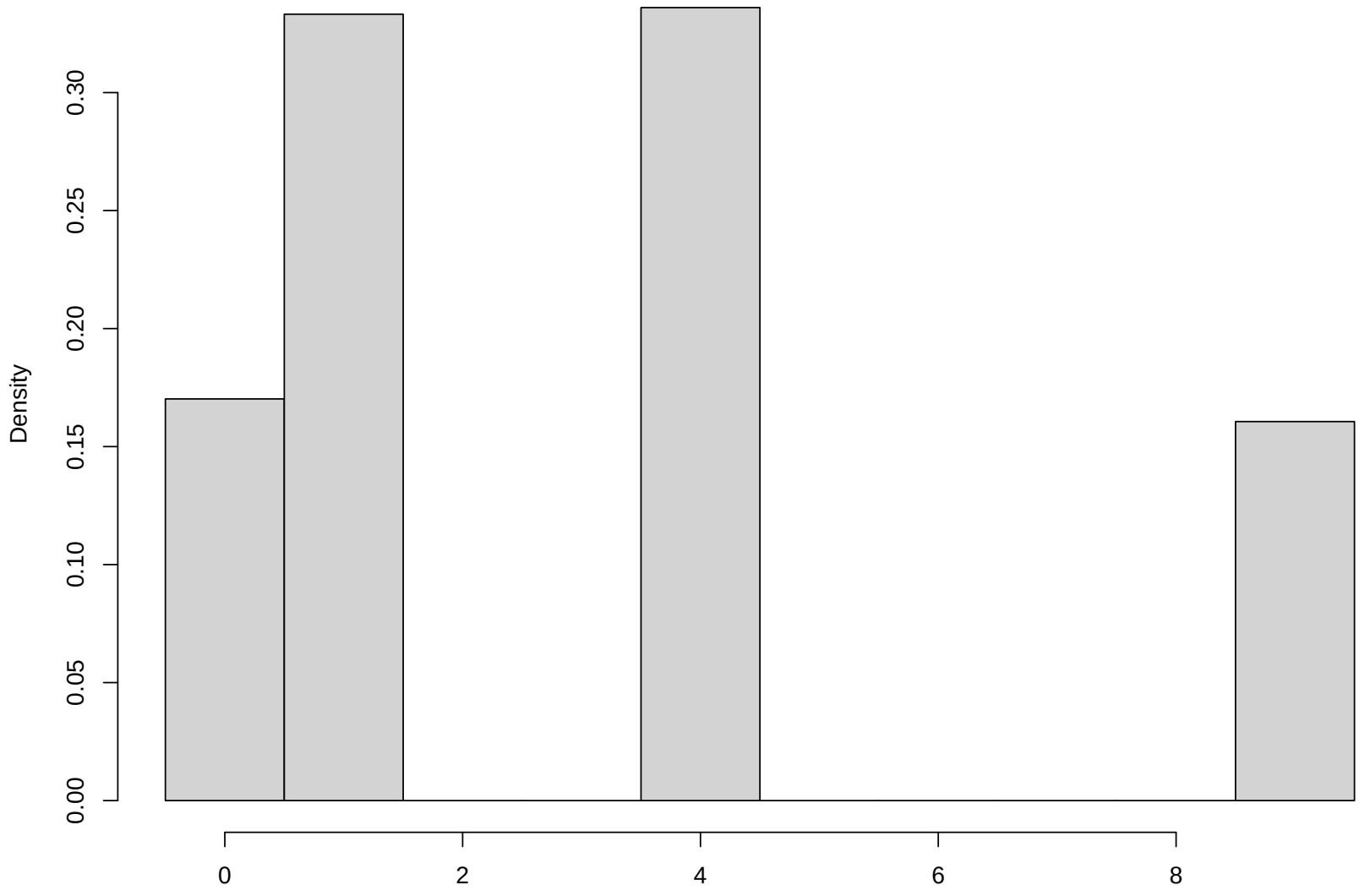
Data Histogram for 1,000 Draws of Y

```
int_hist(sim_y(1000))
```



Data Histogram for 10,000 Draws of Y

```
int_hist(sim_y(10000))
```



Using Simulation to Approximate Probability Histograms

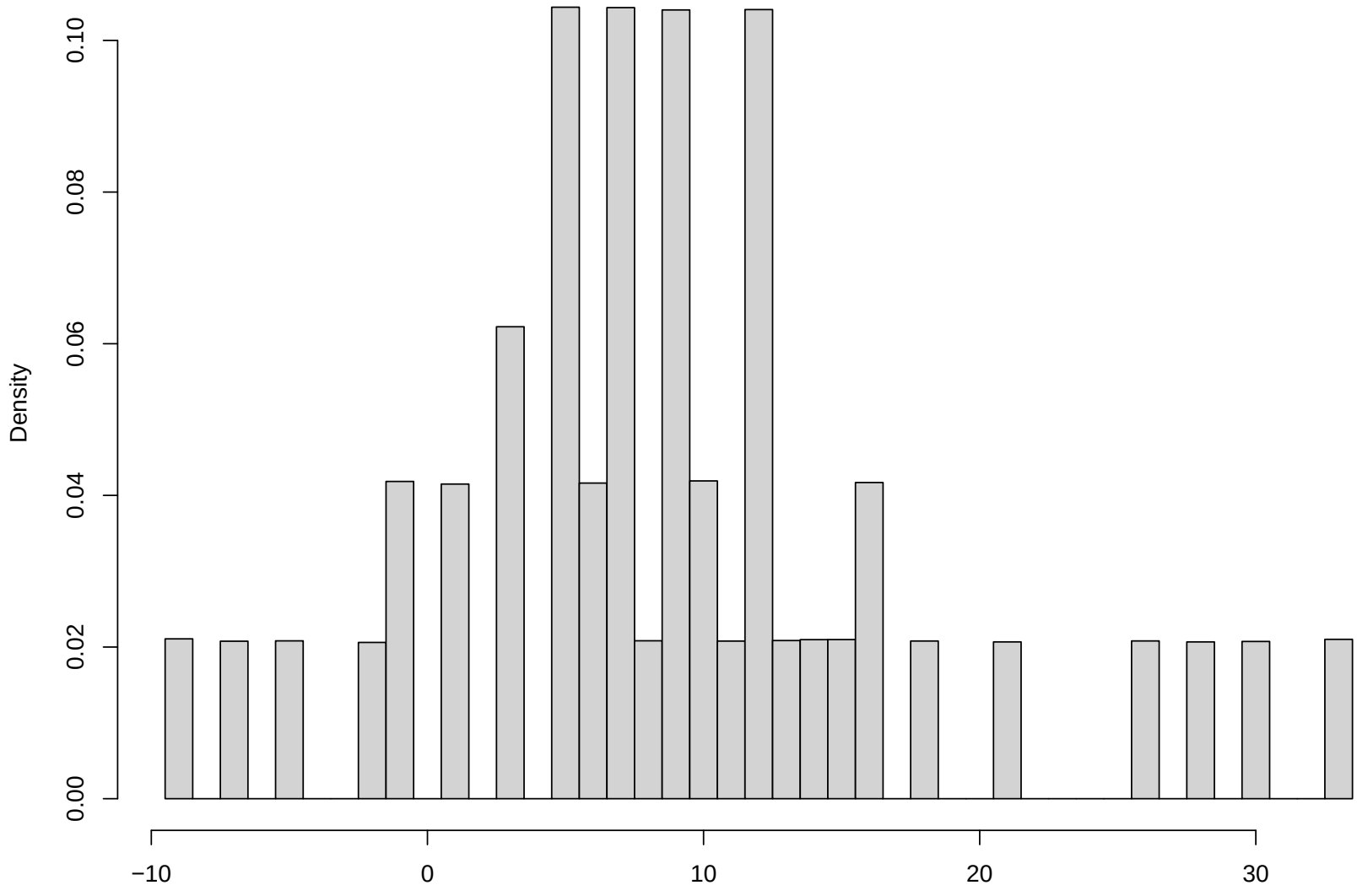
- Suppose box A contains tickets $\{1, 3, 7, -2\}$.
- Suppose box B contains tickets $\{-2, 0, 3\}$.
- Suppose box C contains tickets $\{5, 7, 9, 12\}$.
- Let X, Y, and Z be draws from A, B, and C, respectively.
- Let $U = XY + Z$
- Can we approximate the probability histogram for U by simulation?

Simulating U

```
#1,000,000 draws from A,B,C
X = sample(c(1,3,7,-2),1000000,replace=TRUE)
Y = sample(c(-2,0,3),1000000,replace=TRUE)
Z = sample(c(5,7,9,12),1000000,replace=TRUE)
#Create 1,000,000 values for U
#The operation * multiplies component-wise
U = X*Y+Z
```

Probability Histogram for U

```
int_hist(U)
```



Another Example

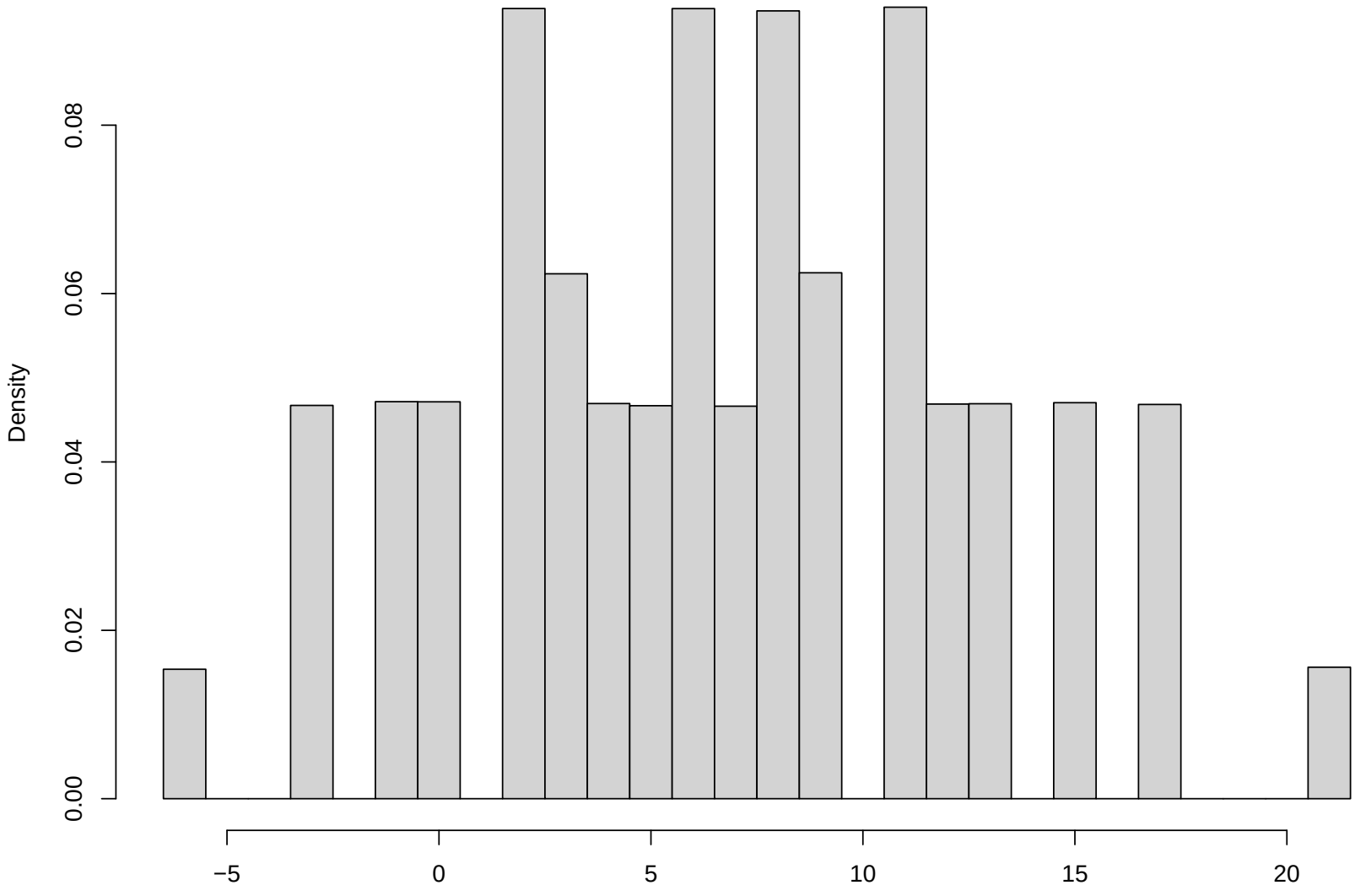
- Suppose box A contains tickets $\{1, 3, 7, -2\}$.
- Let X_1, X_2, and X_3 be three draws with replacement from A.
- Let $S = X_1 + X_2 + X_3$ be the sum.
- What is a probability histogram for S?

Simulating S

```
X1 = sample(c(1,3,7,-2),1000000,replace=TRUE)
X2 = sample(c(1,3,7,-2),1000000,replace=TRUE)
X3 = sample(c(1,3,7,-2),1000000,replace=TRUE)
S = X1+X2+X3
```

Probability Histogram for S

```
int_hist(S)
```



The Central Limit Theorem

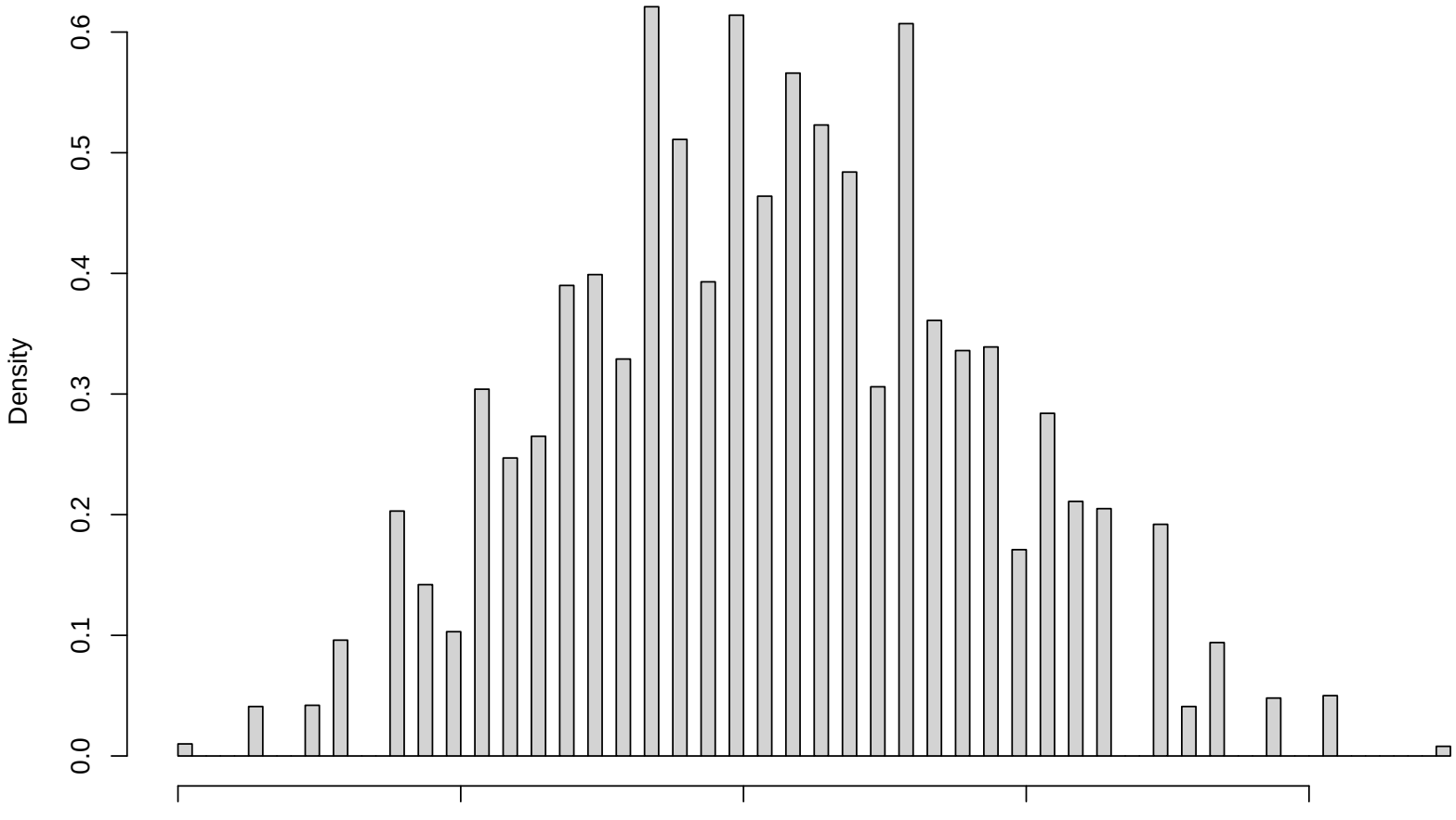
- Suppose we have a sequence of IID random variables X_1, X_2, X_3, ..., X_n.
  - Think draws with replacement from a box.
- The **sample mean** is $\bar{X} = (X_1 + X_2 + X_3 + ... + X_n)/n$.
- The **Central Limit Theorem** states that with more and more draws, the probability histogram of $\bar{X}$ converges to a normal curve.

Simulating a Sample Mean

```
#Returns a sample mean from n draws from a box b
smean = function(b,n) {
  x = sample(b,n,replace=TRUE)
  return(mean(x))
}
#Returns a vector of m sample means
smeans = function(m,n,b) {
  sampmeans = c() #Empty vector
  for (i in 1:m) {
    #Append new sample mean to sampmeans
    sampmeans=c(sampmeans,smean(b,n))
  }
  return(sampmeans)
}
```

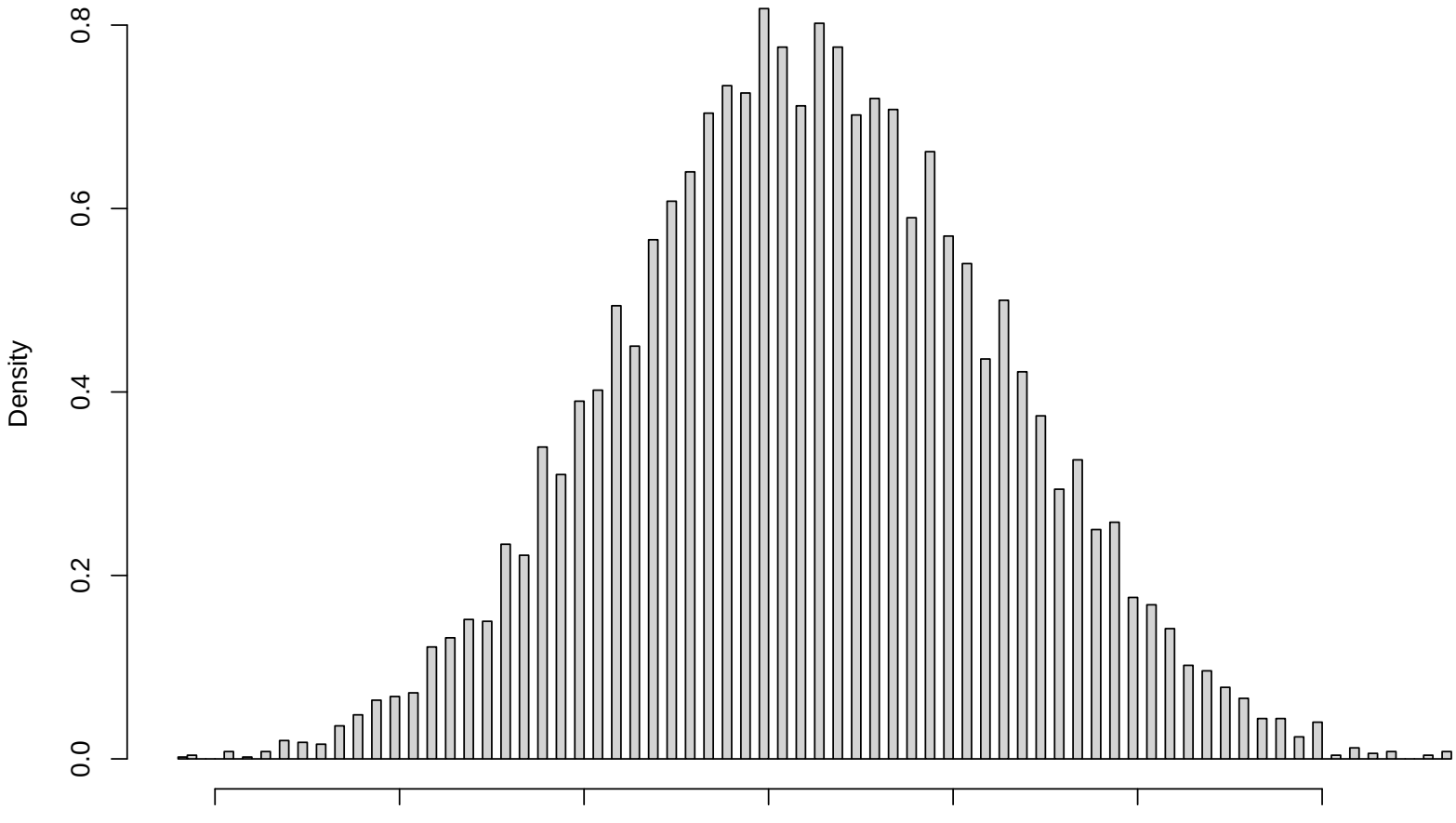
Probability Histogram for $n = 5$

```
X = smeans(10000,5,c(1,3,7,-2))
hist(X,main=NULL,xlab="Sample mean of 5 draws",
     breaks=100,freq=FALSE)
```



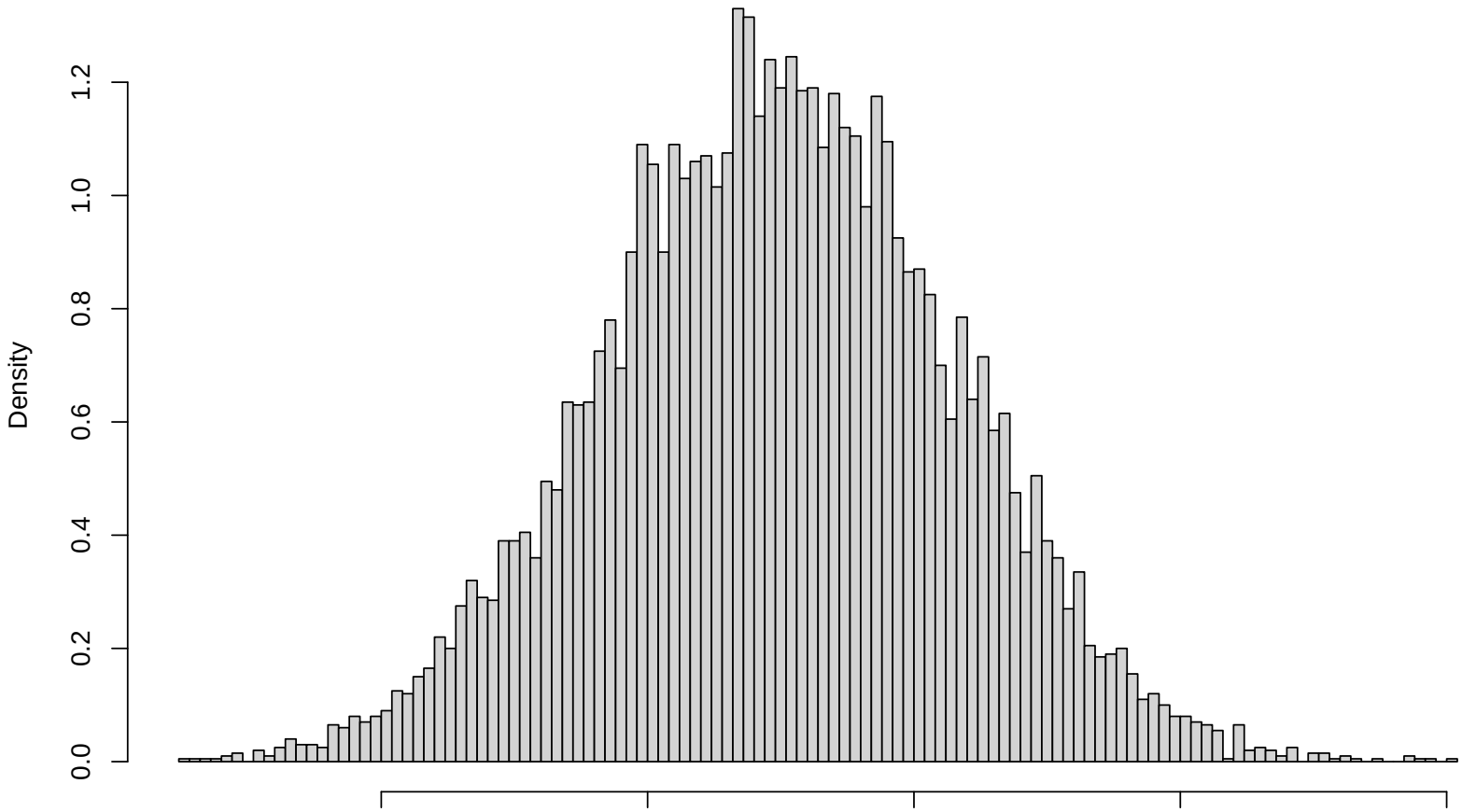
Probability Histogram for $n = 10$

```
X = smeans(10000,10,c(1,3,7,-2))
hist(X,main=NULL,xlab="Sample mean of 10 draws",
     breaks=100,freq=FALSE)
```



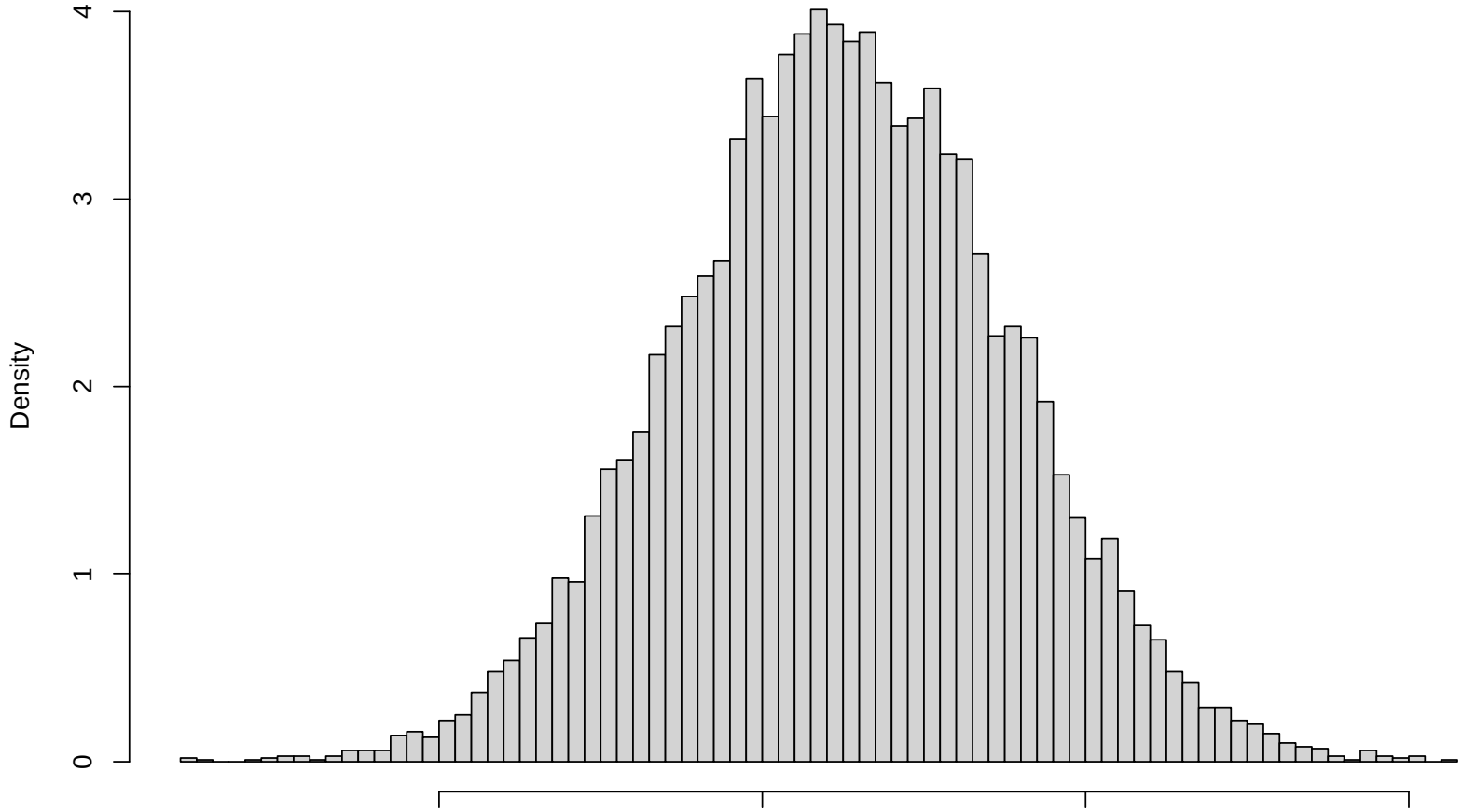
Probability Histogram for $n = 100$

```
X = smeans(10000,100,c(1,3,7,-2))
hist(X,main=NULL,xlab="Sample mean of 100 draws",
     breaks=100,freq=FALSE)
```



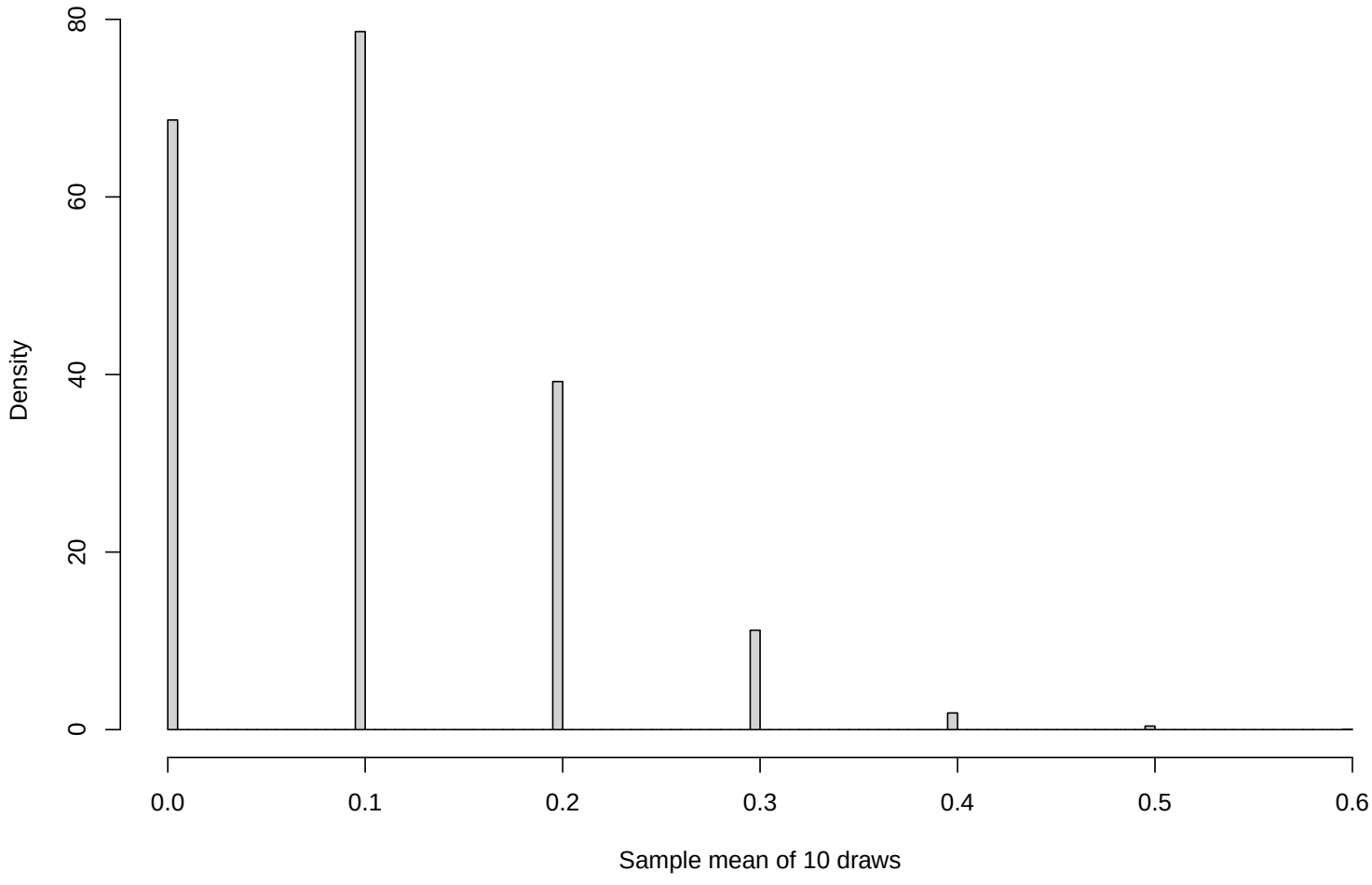
Probability Histogram for $n = 1,000$

```
X = smeans(10000,1000,c(1,3,7,-2))
hist(X,main=NULL,xlab="Sample mean of 1000 draws",
     breaks=100,freq=FALSE)
```



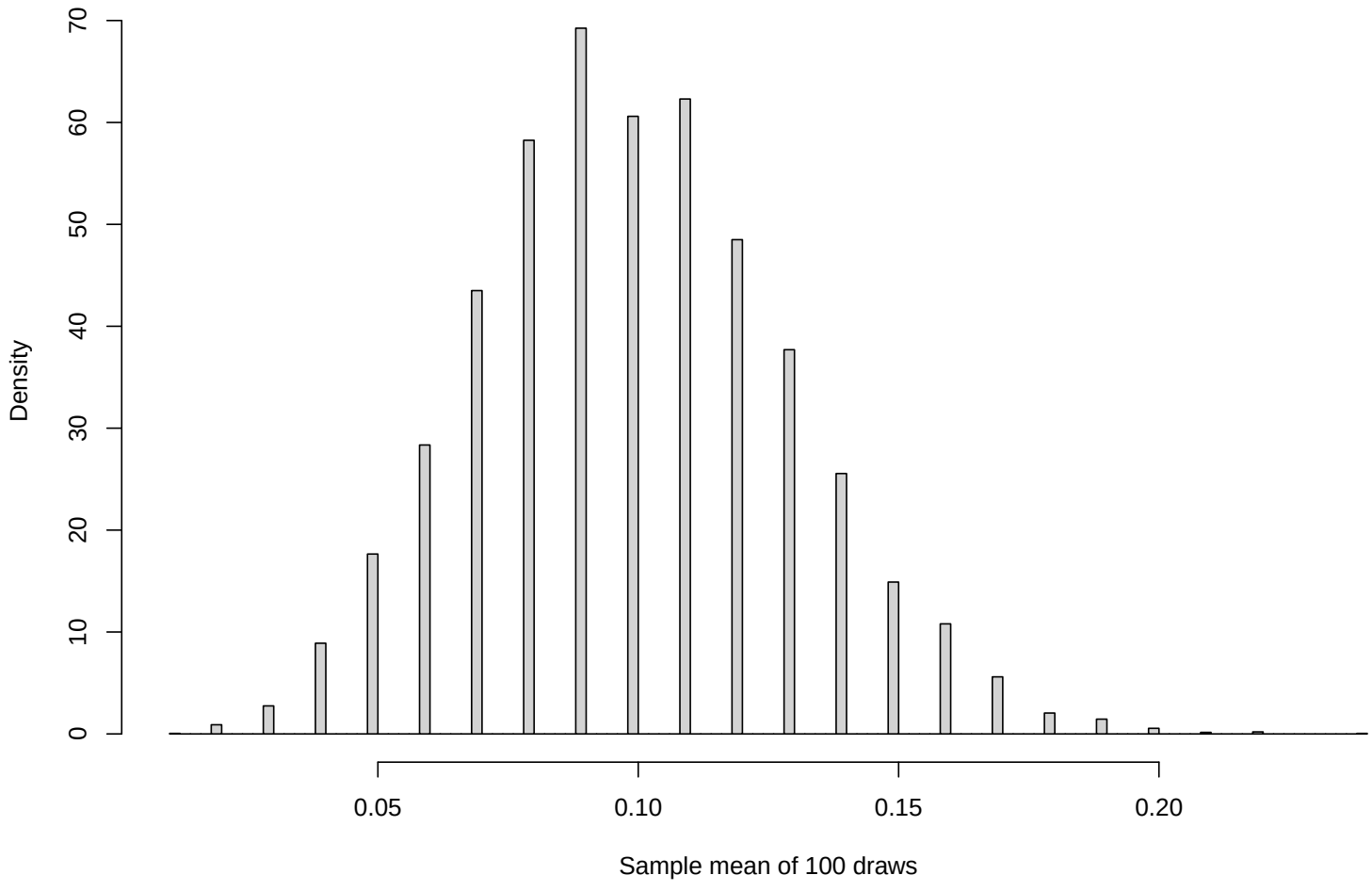
Probability Histogram for Lopsided Box, $n = 10$

- Suppose that the box has 9 0's and 1 1.



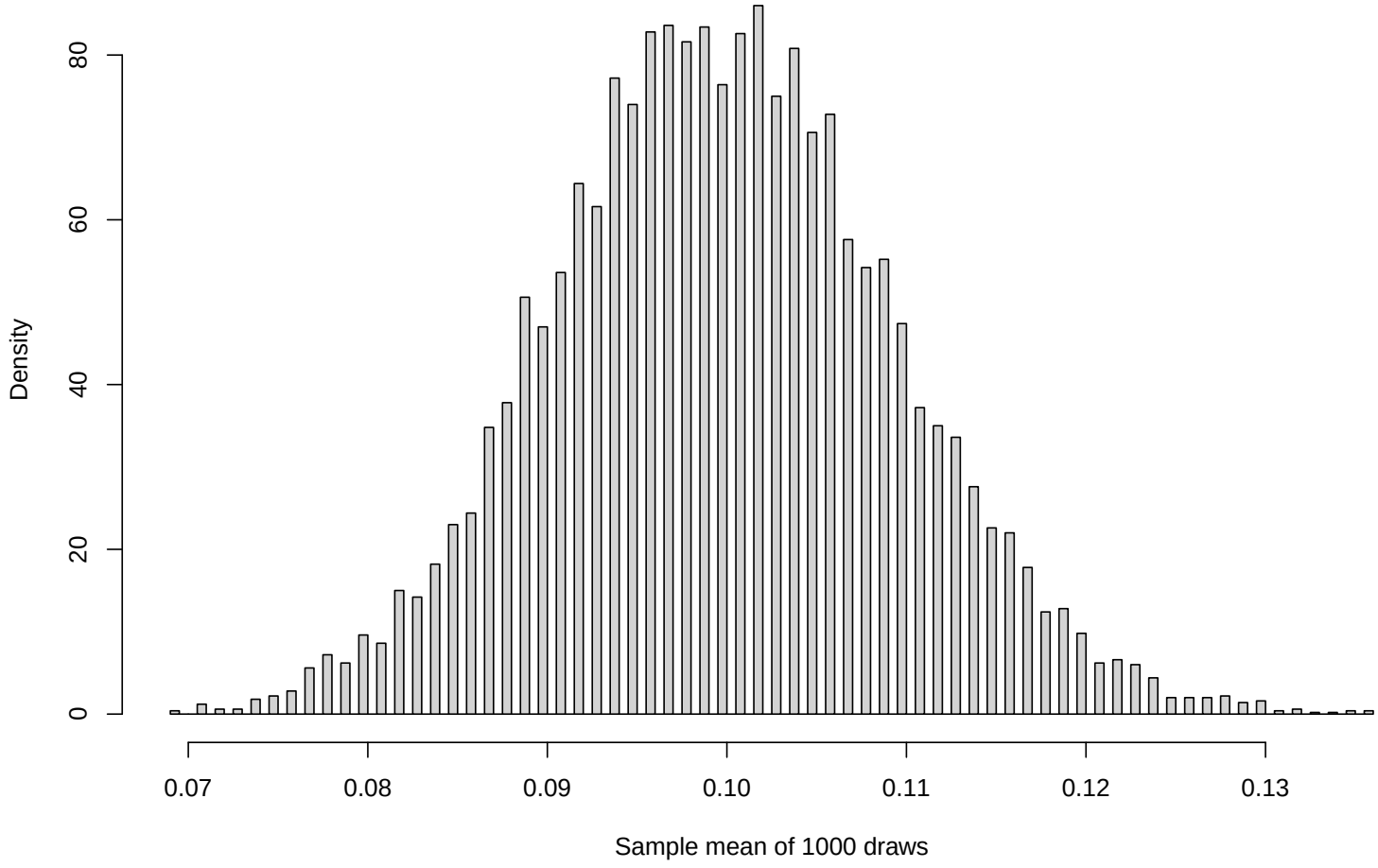
Probability Histogram for Lopsided Box, $n = 100$

- Suppose that the box has 9 0's and 1 1.



Probability Histogram for Lopsided Box, $n = 1000$

- Suppose that the box has 9 0's and 1 1.



Using the Normal Approximation

Simple Example

- A box contains the tickets $\{0, 1\}$.
- We draw from this box 100 times with replacement.
- What is the probability that the sum of the draws will be between 40 and 60 inclusive?

Three Possible Strategies

- We can answer this question in one of three ways:
  - Exactly, using the binomial formula
  - Approximately, using computer simulation
  - Approximately, using the normal approximation

Exact Solution

```
#Probability of 40 to 60 inclusive
print(pbinom(60,100,0.5)-pbinom(39,100,0.5))
```

```
## [1] 0.9647998
```

Computer Simulation

```
#Simulates 1,000,000 binomial random variables
#with 100 draws and probability of 0.5 for each draw
sims = rbinom(1000000,100,0.5)
#The percentage of simulated values between 40 and 60
#inclusive
print(mean(sims>=40 & sims<=60))
```

```
## [1] 0.964722
```

Normal Approximation

- The sum of the draws between 40 and 60 is equivalent to the mean (percentage) being between 0.4 and 0.6.
- The continuity correction gives 39.5(0.395) and 60.5(0.605).
- Expected value of mean $= p = 0.5$.
- Standard error of mean $= \sqrt{\frac{p*(1-p)}{n}} = \sqrt{0.5 * 0.5/100} = 0.05$.
- $z_1 = (0.395 - 0.50)/0.05 = -2.1$.
- $z_2 = (0.605 - 0.50)/0.05 = 2.1$.

Normal Approximation (continued)

```
print(pnorm(2.1)-pnorm(-2.1))
```

```
## [1] 0.9642712
```

A Slightly(!) More Complicated Example

- A box contains the tickets $\{1, 2, 3\}$.
- We draw from this box 100 times with replacement.
- What is the probability that the sum of the draws will be between 190 and 210 inclusive?

Three Possible Strategies

- We can answer this question in one of three ways:
  - Exactly, enumerating all possible outcomes
  - Approximately, using computer simulation
  - Approximately, using the normal approximation

Exact Solution

- There are 3^{100}, or about 1 trillion trillion trillion trillion possible outcomes.
- One possible outcome is 3 3 2 1 3 2 1 3 2 2 1 3 3 3 3 2 3 2 3 3 3 2 1 2 1 2 1 1 2 1 3 1 3 3 3 1 3 1 2 2 1 2 2 3 3 2 2 1 1 1 2 1 3 2 2 1 3 1 3 1 2 2 2 3 1 2 3 3 3 2 2 1 1 3 3 2 2 1 1 2 3 3 3 3 3 1 2 2 1 3 2 3 1 2 2 3 2 3 3, which sums to 208
- How many other possible outcomes sum to 208?
- No simple formula for this
- **An exact solution is not feasible, even using a computer!**
- **There are just too many possibilities to consider.**

Computer Simulation

```
sample_sums = c() #Empty vector
for (i in 1:10000) {
  sample = sample(c(1,2,3),100,replace=TRUE)
  sample_sums = c(sample_sums,sum(sample))
}
#The percentage of sample_sums between 190 and 210
#inclusive
print(mean(sample_sums>=190 & sample_sums<=210))
```

```
## [1] 0.8088
```

Normal Approximation

- Sum between 190 and 210 is equivalent to the mean being between 1.895 and 2.105.
  - This is using the continuity correction.
- Expected value of mean = average of box = 2.
- Standard deviation of box = $(1^2 + 2^2 + 3^2)/3 - 2^2 = 0.8165$.
- Standard error of mean = $SD/\sqrt{100} = 0.081645$.
- $z_1 = (1.895 - 2)/0.08165 = -1.286$.
- $z_2 = (2.105 - 2)/0.08165 = 1.286$.

Normal Approximation Part 2

```
print(pnorm(1.286)-pnorm(-1.286))
```

```
## [1] 0.8015569
```

Exact Solution vs. Computer Simulation vs. Normal Approximation

- Exact solutions are not generally available.
- Computer simulation and the normal approximation can be used in many, many situations in which exact solutions are not possible.
- Historically, the normal approximation was the only option.
  - With modern computers, we now have simulation as well.

Advantages and Disadvantages

- Advantages of simulation:
  - **Can be used even with small samples, when the normal approximation doesn't apply.**
  - Approximation can be made arbitrarily good, depending on computing resources.
  - Easy, even if you forget statistical formulas.
- Advantages of normal approximation:
  - Computationally much faster because for loops can take a long time.
  - Useful for statistical theory because you actually have a formula to work with.

Review Questions

- See Review_questions16.pdf on Canvas